Glenohumeral and Acromioclavicular Osteoarthritis

Background

- Osteoarthritis (OA) is a painful, degenerative condition that involves the deterioration of joint cartilage resulting in inflammation of the joints, pain, and functional impairment. OA is typically caused by wear and tear over time, but it can also be accelerated by traumatic injuries or prior surgery.
- There are two types of OA that occur in the shoulder joints:
  - **Glenohumeral OA:** Arthritis in the shoulder socket that often causes pain during shoulder movements in any direction.
  - **Acromioclavicular OA:** Arthritis in the collar bone that often causes pain when reaching across the body or sleeping directly on the affected shoulder.
- Primary symptoms of OA include joint pain and functional impairment.
- Diagnosis is made through a review of your history, physical exam, and medical imaging by x-ray and MRI.
- Treatment and management of shoulder OA includes stretching and strengthening exercises, regular low-intensity physical activity, and maintaining a healthy body weight. Injections may be used to treat this condition as needed.

Latest Research

- Rehabilitation for shoulder osteoarthritis emphasizes strengthening and stabilizing your "shoulder girdle," the muscles and bones that connect your arm to your torso. These rehab exercises are designed to improve function and should not make your arthritis worse.
- If rehab doesn't help to alleviate pain and improve function, additional treatments may be considered. These include:
  - **Corticosteroid** injections are often referred to as "cortisone shots" and can help reduce inflammation and pain.
  - **Dextrose prolotherapy** involves injecting a solution into your shoulder to help with tissue repair and reduce pain.
  - **Platelet-Rich Plasma (PRP) therapy** involves injecting concentrated platelets from your blood into your shoulder to promote healing and reduce pain.

Rehab Instructions

Range of Motion (ROM) Exercises

Flexion (A):

1. Start by lying down on your back with a stick or cane held in both hands and hold it just above your thighs.
2. Keep your elbows straight and slowly raise the stick over your head.
3. Stop when you feel a slight stretch or discomfort.
4. Hold this position for 5 seconds, then slowly return to the start position.

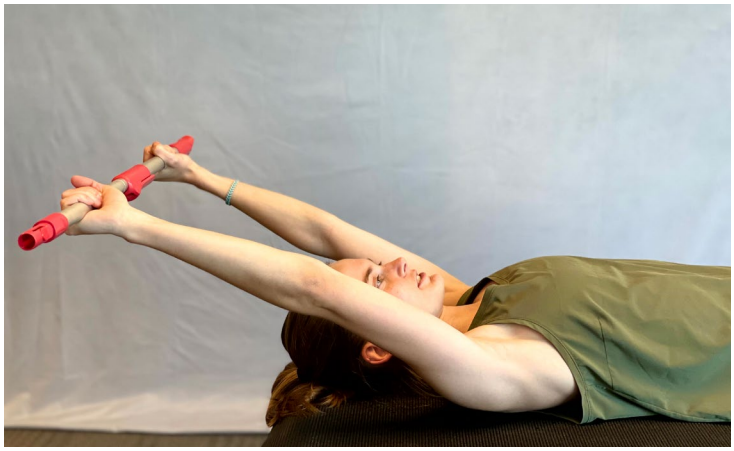
Complete 1 set of 10 reps, 1 x a day.



Abduction (B):

1. Start by lying down on your back and hold one end of the stick with the hand that's on the same side as your affected shoulder.
2. Place your other hand on the opposite end of the stick and gently push the stick to move your affected arm away from your body.
3. Hold this position for 5 seconds, then slowly return to the start position.

Complete 1 set of 10 reps, 1 x a day.



TIPS

- Perform each ROM exercise in a slow and controlled manner.
- Stop the exercise if the pain exceeds a level of 5 out of 10.
- If you experience any instability in your shoulder, stop the exercise and contact your care provider.

Scapular Stabilization

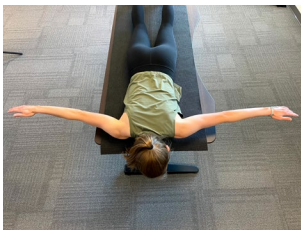
Ts (C)

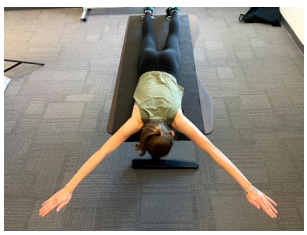
1. Lie face down with your arms straight out at a 90-degree angle to your body and with your palms facing down, forming a "T."
2. Squeeze your shoulder blades together and down as if you're trying to put them into your back pockets.
3. Then, slowly bring your arms up off the floor and towards the ceiling.
4. Hold this position for 5 seconds.

Ts (Y)

1. Lie face down with your arms straight out at about a 45-degree angle to your body with your palms facing down, forming a "Y."
2. Squeeze your shoulder blades together and down as if you're trying to put them into your back pockets.
3. Then, slowly raise your arms up off the floor towards the ceiling.
4. Hold each position for 5 seconds. Perform 2 sets of 10 reps

Complete 2 sets of 10 reps of both C and Y T exercises 1 x a day.





TIPS

- Avoid "hiking up your shoulders." That is, don't bring your shoulders up to your neck.
- You can place a rolled-up towel under your forehead to keep your head off the floor.

Rotator Cuff Strengthening

You'll need a resistance band for these exercises.

Shoulder Scaption (E)

1. Secure the resistance band by putting it under your foot.
2. Hold the resistance band at your side and bring your arm slightly forward.
3. Raise your arm to shoulder height, keeping your elbow straight. You could also use a 1–3-lb hand weight for resistance instead of a resistance band.
4. Hold for 3 seconds.



Internal Rotation (F)

1. Secure a resistance band at the same height as your elbow by closing it in the hinge side of a door such as a bathroom or bedroom door. When you've closed the band in the door, tug on it to test that it's secure.
2. Put a rolled-up towel under your exercising arm. This helps to keep your elbow aligned with your body and ensures proper form. **(F1)**
3. Hold the free end of the resistance band with your exercising hand and position your hand in front of your stomach, putting tension on the resistance band.
4. Slowly pull the band towards your stomach and be careful to not move your elbow. **(F2)**





5. Hold for 3 seconds.

External Rotation (G)

1. Secure a resistance band at the same height as your elbow by closing it in the hinge side of a door such as a bathroom or bedroom door. When you've closed the band in the door tug on it to test that it's secure.
2. Put a rolled-up towel under your exercising arm. This helps to keep your elbow aligned with your body and ensures proper form. **(G1)**
3. Hold the free end of the resistance band with your exercising hand and position your hand in front of your stomach, putting tension on the resistance band.
4. Slowly pull the band *away* from your stomach and be careful to not move your elbow. **(G2)**
5. Hold for 3 seconds.

Complete 2–3 sets of 10 reps of each exercise 1 x a day.





TIPS

- Keep your upper body facing forward, and don't rotate your upper body while completing these exercises.
- Increase band resistance when these exercises get to be easy and pain-free.
- Stepping further away from the door increases the resistance.

References

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